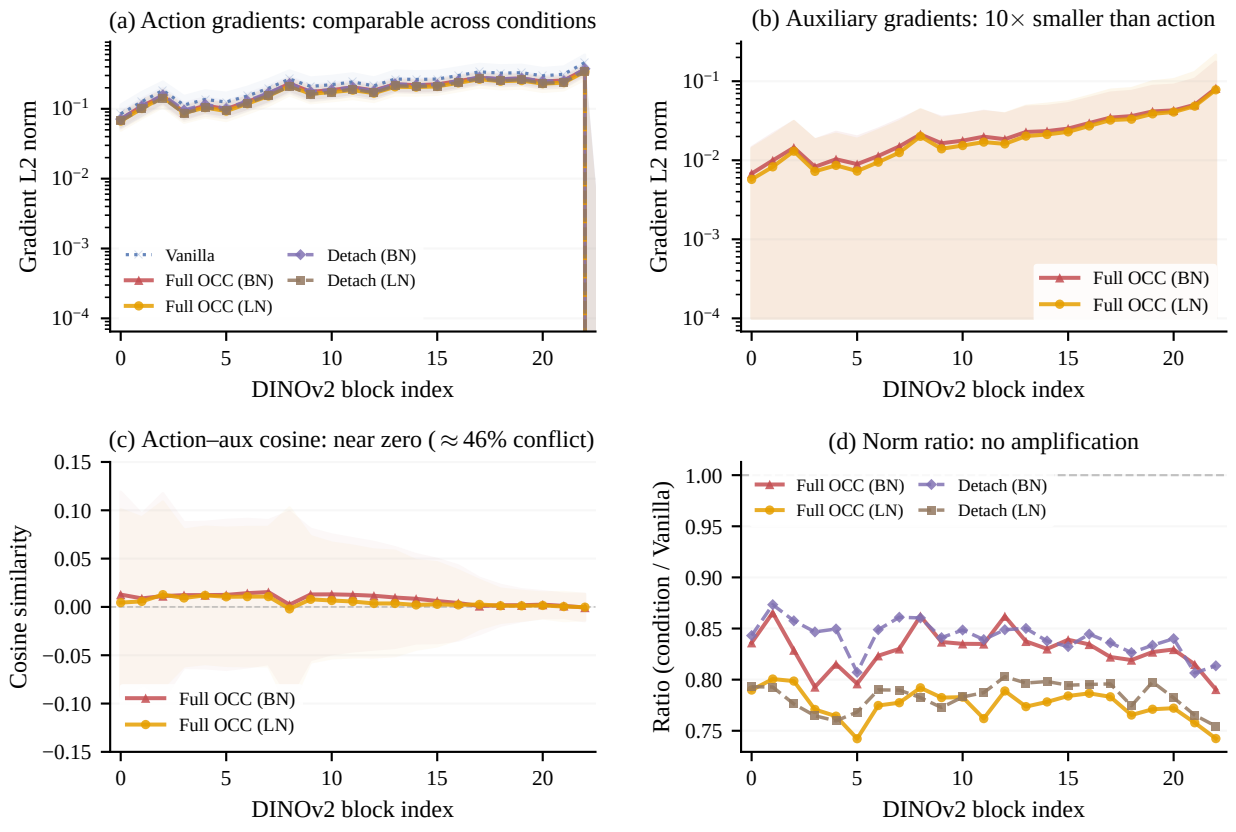
(a) Action gradients: comparable across conditions
Gradient L2 norm
DINOv2 block index
Vanilla
Full OCC (BN)
Full OCC (LN)
Detach (BN)
Detach (LN)
(b) Auxiliary gradients: 10× smaller than action
Gradient L2 norm
DINOv2 block index
Full OCC (BN)
Full OCC (LN)
(c) Action–aux cosine: near zero (≈ 46% conflict)
Cosine similarity
DINOv2 block index
Full OCC (BN)
Full OCC (LN)
(d) Norm ratio: no amplification
Ratio (condition / Vanilla)
DINOv2 block index
Full OCC (BN)
Full OCC (LN)
Detach (BN)
Detach (LN)